

nated from an attached PC. A virus can be created whose payload is triggered by executing on a PC that is attached to the target network. This is not hard with a little inside information about the configuration of the network. The propitiator would then install the virus at all of the local Universities in the hope that someone working at the installation is taking a night class or that one of their children will unknowingly infect a common usage home computer. At that point, the virus has a good chance of entering the target network. This is a well known vector and is enhanced because the virus will not reveal itself. Once on the target system, the PC virus will act like a dropper releasing a Unix virus into the backbone. The payload virus may be necessary because many Unix backbone systems are not PC compatible. The Unix virus payload can then install a back door which can be remotely directed. In addition, the virus can create a covert channel by making use of messenger viruses. While the use of messenger viruses are slow and have low bandwidth, they are bidirectional and can be used for command and control of more complex attacks.

Conclusion

The problem of attack software targeted against Unix systems will continue to grow. Viruses may become more prevalent because they provide all of the benefits of other forms of attack, while having few drawbacks. Trans-platform viruses may become common as an effective attack. All of the methods currently used in creating MS-DOS viruses can be ported to Unix. This includes the creation of automated CAD/CAM virus tools, stealth, polymorphism and armour. The future of viruses on Unix is already hinted at by the wide spread use of Bots and Kill-bots (slang term referring to software robots). These programs are able to move from system to system performing their function. Using a Bot as a dropper or creating a virus that includes bot like capability is simple. With the advent of global networks, the edge between viruses, bots, worms and Trojans will blur. Attacks will be created that use abilities from all of these forms and others to be developed. There have already been cases where people have used audit tools such as COPS